How can we tell when a liquid is a mixture?

1 Use the words below to fill the gaps in these sentences. You may need to use a word more than once.

dissolve	mixtures	soluble	insoluble	solutions	filtering

If you take sugar or salt and _____ it in water you can't see it any more.

Sugar is _____ which means that it will dissolve. Sand is _____ which means that it won't dissolve.

Solutions are _____, because you can get back the substance you dissolved by heating the solution until the liquid has evaporated leaving the substance behind.

If a substance in a mixture is insoluble you can sometimes separate it by _____.

2 Complete the table.

Solid	Liquid	Does it dissolve?	Name of solute
salt	water	yes	
sugar	water	yes	
copper sulphate	water	yes	
sand	water	no	

ɑ I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme

How much salt is in this rock?

1 Use the words below to fill the gaps in these sentences. You may need to use some words more than once and other words not at all.

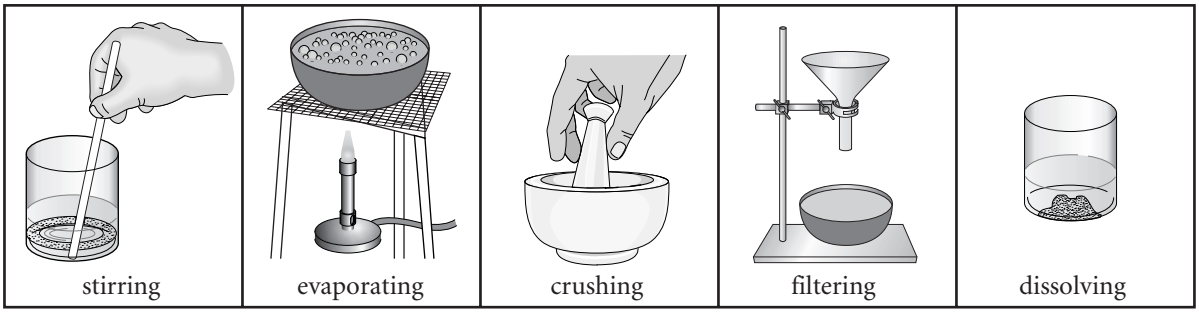
salt	**impurities**	**dissolve**	**crush**	**sodium**
crystals	**filter**	**rock salt**		

Salt is a chemical whose name is sodium chloride. This means it is made of two chemicals, _____ and chlorine.

Pure _____ with a few other materials such as sand mixed in is called _____. As we don't want these extras and they make the salt impure, we call them _____.

To get the salt out of rock salt we _____ it and add water to _____ the salt. Then we _____ the mixture to get rid of the undissolved impurities. Finally, we get _____ of pure salt from the salt solution by evaporation.

2 Put the diagrams below in order to show how we can get pure salt from rock salt.



ɑ I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme

What happens to the solute during dissolving?

1 Use the words below to fill the gaps in these sentences. You may need to use a word more than once.

120 g	**conserved**	**crystal**	**dissolving**
disappear	**particles**	**solute**	**solution**

When you dissolve 20 g of _____ in 100 g of water else you get a solution of _____. The mass of the solute and mass of the liquid is always the same as the total mass of the _____. We say mass is _____.

A _____ of solid salt will break up into very small pieces that are surrounded by water _____. The pieces are too small to see, so the salt seems to _____.

2 Write down the total mass of solution that will be made when the following substances are dissolved.

a 10 g of copper sulphate in 90 g of water _____

b 20 g of copper sulphate in 100 g of water _____

c 15 g of copper sulphate in 100 g of water _____

d 12 g of copper sulphate in 200 g of water _____

e 25 g of copper sulphate in 190 g of water _____

f 200 g of copper sulphate in 1500 g of water _____

q I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme

How do we get the solvent back from a mixture?

1 Use the words below to fill the gaps in these sentences. You may need to use a word more than once.

liquid	**solvent**	**distillation**	**mixture**
solute	**condenses**	**evaporates**	

A solvent is a _____ that will dissolve a _____ .

If you have sea water but no fresh water you might want to get

the _____ out of the sea water.

The process which is used to get the solvent out of a _____ is

called _____ . First the mixture is heated so that the

solvent _____ , then the evaporated solvent is cooled down so

that it _____ back into a liquid again and can be collected.

2 Put the steps below into the correct order to get pure water from sea water by distillation.

A the vapour condenses.

B Heat the sea water.

C Collect the liquid water in a cup.

D Cool the water vapour down.

E The water evaporates leaving any solutes behind.

q I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme

Using chromatography to separate mixtures

1 Use the words below to fill the gaps in these sentences. You may need to use a word more than once.

separated	**solvent**	**filter**	**paper**
solute	**mixture**	**chromatogram**	

Chromatography is another way of finding out whether a liquid is a pure substance or a _____ of substances. The _____ carries the solute particles along as it flows over the _____ paper. Some _____ particles aren't carried as far because they stick to the _____ better than others. This means the bands of colour are _____ from each other.

The result is a _____, or ''colour picture''.

2 Which mixtures can be separated by which methods? Join the boxes.

Mixture	**Method**
Sand and water	Crystallisation
A mixture of soluble dyes	Filtering
Salt and water	Chromatography
Sugar and ethanol	Evaporation

ɑ I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme

What affects dissolving?

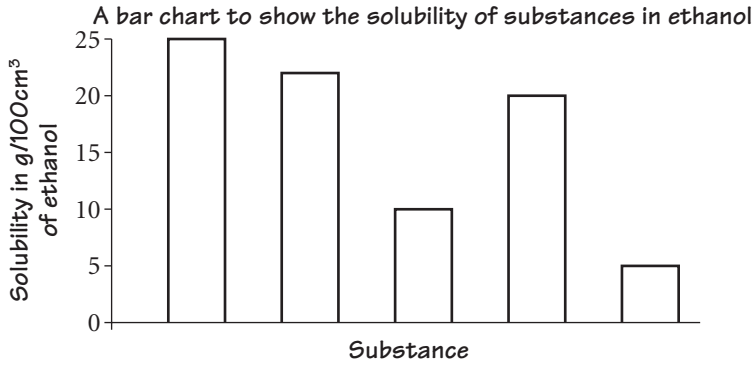
1. Use the words below to fill the gaps in these sentences. You may need to use a word more than once.

saturated	dissolve	stir	solvent
solute	temperature	solution	

A solution that has dissolved so much solute that it can't _____ any more is called a _____ solution. You can tell when a _____ is saturated because if you add more solute it falls to the bottom and won't dissolve no matter how long you _____ it for. Different solutes can dissolve in the same _____, but different amounts of each solute will dissolve. Some solutes dissolve in one solvent and some will dissolve in others. Many solutes will dissolve in more than one _____ .

How much _____ dissolves in a solvent depends on the _____ of the solvent.

2 Joanna measured how much of different substances dissolved in ethanol, and drew the bar chart below to show her results, but she forgot to label it.
Draw a line from each box to the correct bar on the bar chart.



substance A	substance B	substance C	substance D	substance E
20 g per 100 cm^3	10 g per 100 cm^3	5 g per 100 cm^3	25 g per 100 cm^3	22 g per 100 cm^3

q I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme

Why are fuels useful?

Use the words in the box below to fill the gaps in these sentences. You may need to use a word more than once.

°C	heat	gas	temperature	fuel	coal	light
thermometer	oil	hot	transfer			

1 a A substance which is burnt to release energy is called a _____.

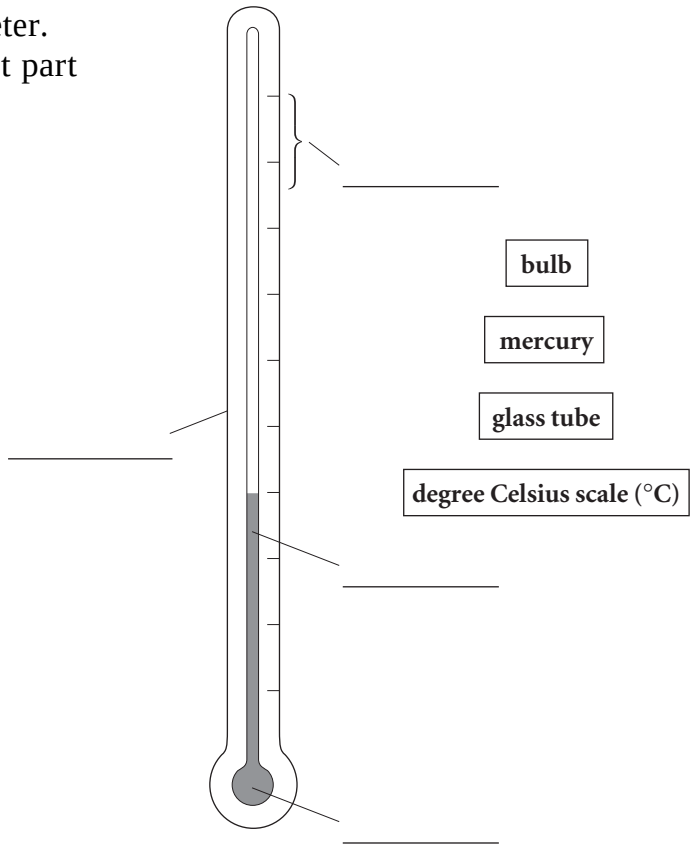
_____, _____ and _____

are examples of everyday fuels used to generate electricity.

b Energy can be found in many forms and it can _____ from one form to another. _____ and _____ are examples of energy which are released when a fuel is burnt.

c A _____ is a piece of equipment used to measure how _____ an object is, or its _____. The unit used to measure the temperature of an object is _____.

2 Below is a diagram of a thermometer. Match the correct label to the right part of the thermometer.



ɑ I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme